AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS X EXAMINATION

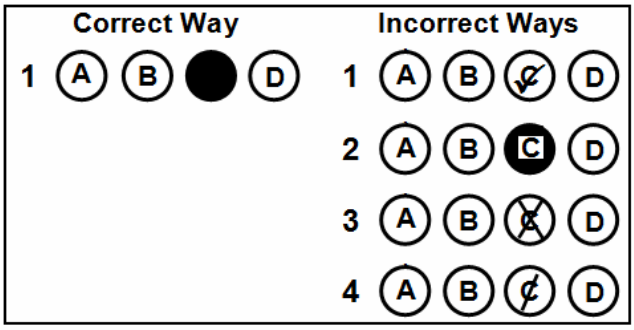
MAY 2014

General Mathematics Paper I

Time: 40 minutes Marks: 30

INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 30 only.
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. You may use a simple calculator if you wish.

1. Which of the following is a polynomial?

   A. $3x$
   B. $3\sqrt{x}$
   C. $\frac{3}{x}$
   D. $3x^{\frac{1}{3}}+2$

2. $\frac{x-b}{x^2-b^2}$ is equal to

   A. $x+b$
   B. $x-b$
   C. $\frac{1}{x-b}$
   D. $\frac{1}{x+b}$

3. If $P(x)=x^3-x^2+2x-2$, then $P(-1)$ is equal to

   A. 0
   B. -2
   C. -4
   D. -6

4. On complete factorization of $2x+3y+3x+2y$, we get

   A. $(x+y)\times 3$
   B. $(x+y)\times 2$
   C. $(x+y)\times 5$
   D. $(x+y)\times 6$

5. When x^3-2x+5 is divided by $x-1$, the remainder is

   A. 4
   B. 5
   C. 6
   D. 7

6. On factorization of $3x+6y-9z$, we get

   A. $3(x+2y-3z)$
   B. $3(x+3y-6z)$
   C. $3(x+6y-9z)$
   D. $3(x-2y+3z)$

7. $\frac{3x-4}{2}-\frac{6x-8}{4}$ is equal to

A. $\frac{12x-6}{4}$

B. $\frac{6x-12}{4}$

C. -4

D. 0

8. $\frac{(3x+6)^2}{x+2}$ is equal to

A. $3(x+2)$

B. $9(x+2)$

C. $3(x+2)^2$

D. $9(x+2)^2$

9. The highest common factor (HCF) of $a + b$ and $a^2+2ab+b^2$ is

A. a^2+b^2

B. a^3+b^3

C. $(a+b)^3$

D. $a+b$

10. The solution set of $ax+a=a$ is

[**Note**: $a \neq 0$]

A. $\{0\}$

B. $\{2\}$

C. $\{a\}$

D. $\{2a\}$

11. The solution set of $\frac{\sqrt{x+2}}{2}=0$ is equal to

A. $\{2\}$

B. $\{0\}$

C. $\{-2\}$

D. $\{-2,2\}$

12. If $3x - 6 = 24$, then x is equal to

A. 6
B. 10
C. 15
D. 25

13. If the solution set of the equation $x^2 - x + p = 0$ is $\{-2,3\}$, then the value of p is

A. -6
B. -1
C. 1
D. 6

14. The solution of quadratic equation $x^2 + 2x = 0$ is

A. $\{-2\}$
B. $\{2\}$
C. $\{0,-2\}$
D. $\{0,2\}$

15. $\begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ is equal to

A. $\begin{bmatrix} 2 & 1 \\ 3 & 2 \end{bmatrix}$
B. $\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$
C. $\begin{bmatrix} 3 & 1 \\ 3 & 3 \end{bmatrix}$
D. $\begin{bmatrix} 3 & 2 \\ 4 & 3 \end{bmatrix}$

16. The multiplicative identity of every 2×2 matrix is

A. $\begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix}$
B. $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$
C. $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$
D. $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

17. The determinant of $\begin{bmatrix} -a & b \\ -a & -b \end{bmatrix}$ is

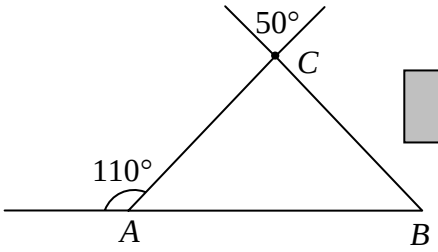
A. 0
B. $-2ab$
C. $2ab$
D. $a^2 - b^2$

18. If $A = \begin{bmatrix} 3 & 4 \\ a & 8 \end{bmatrix}$ and $|A| = 0$, then a is equal to

A. 6
B. -6
C. 7
D. -7

19. In the given triangle, $m\angle B$ is equal to

A. 40°
B. 50°
C. 60°
D. 70°

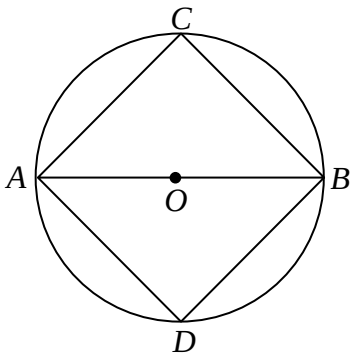


Not to Scale

20. Which of the following options is CORRECT for the given figure?

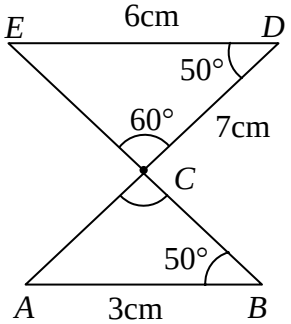
I. $m\angle ACB = m\angle AOB$
II. $m\angle ACB = \frac{1}{2} m\angle AOB$
III. $m\angle ACB = m\angle ADB$

A. I only
B. II only
C. I & III
D. II and III



21. In the given figure, $m\overline{BC}$ is equal to

A. 3 cm
B. 3.5 cm
C. 4 cm
D. 6 cm



Not to Scale

22. Concentric circles have the same

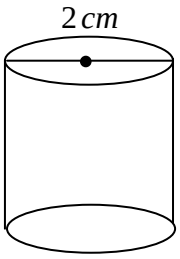
A. centres.
B. radii.
C. areas.
D. tangents.

23. The line segment which joins the midpoint of a side to the opposite vertex is known as an/a

A. angle bisector.
B. altitudes.
C. perpendicular bisector.
D. median.

24. The volume of the given can is

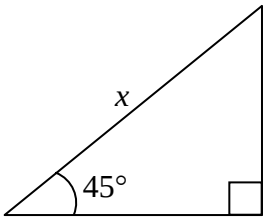
A. $\frac{5}{2}\pi\, cm^3$
B. $5\pi\, cm^3$
C. $10\pi\, cm^3$
D. $15\pi\, cm^3$



Not to Scale

25. For the given right angled triangle the value of x is equal to

A. $2\, cm$
B. $2\sqrt{2}\, cm$
C. $4\, cm$
D. $4\sqrt{2}\, cm$



Not to Scale

26. The area of a square is x and if all its sides are doubled in measure, then the area will become

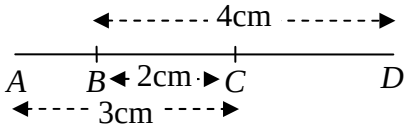
A. $2x$
B. $2x^2$
C. $4x$
D. $4x^2$

27. If the radius of a cylinder is $1\, cm$ and height is $6\, cm$, then the volume of a cone having the same height and radius is

A. $6\pi\, cm^3$
B. $3\pi\, cm^3$
C. $2\pi\, cm^3$
D. $\frac{1}{3}\pi\, cm^3$

28. In the figure *A, B, C* and *D* are collinear points. If $m\overline{AC} = 3cm,\ m\overline{BD} = 4cm$ and $m\overline{BC} = 2cm$, then $m\overline{AD}$ is

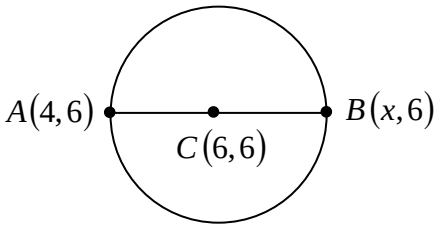
A. 5 cm
B. 6 cm
C. 7 cm
D. 8 cm



Not to Scale

29. If $A(4,6)$ and $B(x,6)$ are the end points of a diameter and $C(6,6)$ is the centre of the circle as shown in the given diagram, then *x* is equal to

A. 4
B. 6
C. 8
D. 10



Not to Scale

30. The distance between the points $(-3,0)$ and $(0,-1)$ is

A. 8
B. $\sqrt{8}$
C. 10
D. $\sqrt{10}$

END OF PAPER

Please use this page for rough work